Depth-Conditioned Feature Compositing for Unsupervised Monocular Panoptic Segmentation

Composing Orthogonal Depth-Semantic Interventions for Pseudo-Label Ceiling Elevation

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Abstract. Unsupervised panoptic segmentation seeks to partition images into semantic regions and object instances without any manual annotation—a task that demands both categorical understanding and individual object separation from unlabeled data alone. The state of the art, CUPS (CVPR 2025), achieves 27.8% PQ by leveraging stereo video sequences and optical flow for multi-view consistency—geometric signals that are unavailable in standard monocular imagery. We show that this coupling is unnecessary: a fully monocular pipeline can not only match but substantially exceed the stereo-dependent state of the art. Our approach rests on the observation that semantic grouping and instance separation admit a natural decomposition into complementary frozen foundation-model signals—appearance features for categories, geometric depth for boundaries. A simple composition of monocular depth estimation (Depth Anything v3) and self-supervised semantic clustering (CAUSE-TR, k=80) generates panoptic pseudo-labels achieving 27.87% PQ on Cityscapes, matching CUPS while using strictly less data. EMA self-training with DINOv3 ViT-B/16 pushes this to 32.76% PQ—but the pipeline then plateaus: depth splits over-fragment and semantic clusters lack geometric awareness. We diagnose these failures as structurally orthogonal error modes and introduce lightweight interventions that compose near-additively. A Depth-Conditioned Feature Adapter (DCFA, 40K parameters) improves semantic clustering by +2.60 mIoU. A Semantic-Instance Mutual Consistency Filter (SIMCF-ABC) improves thing-class quality by +1.48 $\mathrm{PQ_{th}}$. The combined gain amplifies through EMA self-training to **35.83% PQ—+8.0 above CUPS** and **36.26% $\mathbf{PQ_{th}}$—+8.9 above CUPS**. Zero-shot transfer to Mapillary Vistas (39.19% PQ, +3.36 above in-domain) confirms domain invariance.

Keywords: unsupervised panoptic segmentation, monocular depth estimation, DINO, pseudo-label generation, self-training, cross-modal fusion

1. Methodology

The central design challenge in unsupervised panoptic segmentation is that semantic grouping and instance separation demand complementary signals—appearance for categorical understanding, geometry for object boundaries—yet naïve composition of frozen foundation-model outputs leaves three structurally orthogonal error modes unaddressed. Our methodology is built around diagnosing these modes and demonstrating that lightweight interventions targeting each compose near-additively, a property that both validates the orthogonality claim and produces the final result.

Stage 1: composing the baseline. We begin with the simplest possible composition of two frozen foundation models: Depth Anything v3 for monocular depth estimation, and CAUSE-TR applied to frozen DINOv3 ViT-B/16 patch features for semantic clustering. The depth stream extracts instance boundaries via Sobel edge detection followed by connected component analysis, while the semantic stream clusters features via k-means. This baseline already generates pseudo-labels achieving 27.87% PQ on Cityscapes—matching CUPS (27.8% PQ) which requires stereo video and optical flow—suggesting that monocular geometry is sufficient for instance separation when the depth estimator is strong.

The choice of k=80 for k-means clustering is not arbitrary. At k=27, the Cityscapes class count, the clustering suffers from centroid collapse: fourteen of twenty-seven centroids attract no pixels, and seven evaluation classes—fence, pole, traffic light, traffic sign, rider, train, motorcycle—receive zero IoU. This collapse occurs because DINOv3 features on urban scenes are not uniformly distributed: classes like road and building dominate the pixel mass, attracting multiple centroids, while rare classes lack sufficient feature density to sustain dedicated clusters. Overclustering to k=80 redistributes centroid mass, allowing rare classes to attract dedicated clusters while semantically similar frequent classes split into sub-clusters that Hungarian matching later consolidates. Hungarian matching is applied on the global confusion matrix rather than per-image, ensuring consistent cluster-to-class mapping across the dataset; unmatched clusters are assigned via argmax to the class with highest marginal overlap.

The instance stream faces a parallel design question: given a depth map, how should one extract instances? We systematically evaluated five splitting algorithms—Sobel+CC, Canny, watershed, multiscale Sobel, and Mumford-Shah spectral clustering—across four depth estimators. The striking finding is that all algorithms yield nearly identical $\mathrm{PQ}_{\mathrm{th}}$ when depth quality is held constant, and all scale monotonically with depth quality regardless of algorithmic sophistication. Mumford-Shah achieves only +19.9% over Sobel+CC (23.27% vs 19.41% $\mathrm{PQ}_{\mathrm{th}}$ with SPIdepth) while requiring five-hundred times more computation (19.75s vs 0.04s per image); when Depth Anything v3 replaces SPIdepth, even simple Sobel+CC reaches 20.90% $\mathrm{PQ}_{\mathrm{th}}$. This establishes a key principle that guides our design: investment in better monocular depth estimation yields larger returns than investment in sophisticated post-processing. We therefore use Sobel+CC for speed, and direct architectural effort toward DCFA (Section **??**), which targets the feature level rather than the splitting level.

Separating thing classes from stuff classes is accomplished without ground-truth labels by computing, for each semantic class k, the ratio of connected components that survive after masking depth-edge pixels:

$$R_{\mathrm{split}}(k) = \frac{\mathbb{E}_I\big[\mathrm{CC}(M_k \cap \neg\mathcal{E}_{\mathrm{depth}})\big]}{\mathbb{E}_I\big[\mathrm{CC}(M_k)\big]} \tag{1}$$

where M_k is the semantic mask for class k, $\mathcal{E}_{\mathrm{depth}}$ marks depth-edge pixels, and $\mathrm{CC}(\cdot)$ counts connected components. Classes with high R_{split} fragment extensively across depth edges and are classified as things; low-split classes form coherent regions and are classified as stuff.

Raw fusion of semantic and instance masks produces panoptic pseudo-labels that achieve 27.87% PQ on Cityscapes, matching CUPS—but the composition leaves three persistent error modes that prevent further progress.

The three orthogonal error modes. The baseline plateaus because its constituent signals carry independent, non-interacting errors. First, *feature-level errors*: frozen DINOv3 features encode appearance semantics but are trained entirely on two-dimensional images, making them blind to scene geometry. A car hood and the road beneath it may share similar texture—dark, smooth, reflective—and thus cluster together in feature space, despite being at different depths and belonging to different semantic categories. Second, *geometric-level errors*: monocular depth provides natural instance boundaries, but raw splitting produces catastrophic over-fragmentation ($\sim$44 instances per image, 50.7% stuff contamination), because depth discontinuities alone cannot distinguish between a single large object and adjacent background surfaces. Third, *label-level errors*: merged semantic and instance streams suffer cross-modal inconsistency—semantic regions over-spill into geometrically distinct objects, and depth instances fragment semantically coherent objects.

These three failures are structurally orthogonal: they occur at different stages of the pipeline, and improving one does not ameliorate the others. This orthogonality is not merely a conceptual observation; it is an empirical property that manifests as near-additive composition of interventions targeting each mode.

DCFA: conditioning features on geometry without corruption. The feature-level error calls for a module that injects geometric awareness into frozen semantic features without disturbing the clustering manifold that CAUSE-TR relies upon. Fine-tuning the DINOv3 backbone on depth prediction would adapt features to geometry, but early experiments showed that even lightweight backbone adapters degrade downstream k-means clustering by collapsing inter-class margins, costing approximately five PQ. The semantic features are already excellent; what they need is a lightweight geometric cue that does not corrupt the learned manifold.

We introduce the Depth-Conditioned Feature Adapter (DCFA), a forty-thousand-parameter module that modulates frozen DINOv3 patch features with sinusoidally-encoded monocular depth. Monocular depth values are continuous and unbounded; sinusoidal positional encoding with octave-spaced frequencies provides a multi-resolution representation that captures both nearby and distant depth relationships:

$$\gamma(d) = \Big[\sin(2^{\ell}\pi d),\, \cos(2^{\ell}\pi d)\Big]_{\ell=0}^{L-1} \tag{2}$$

The periodic nature of sinusoidal encoding ensures that small depth differences near the camera—where depth estimation is most accurate—receive high-frequency discrimination, while large depth differences in the background receive low-frequency coarse discrimination.

The depth encoding is concatenated with frozen DINOv3 patch features and fused via an MLP with a skip connection:

$$f_{\mathrm{adapt}} = f_{\mathrm{dino}} + \mathrm{MLP}\big([f_{\mathrm{dino}},\, \gamma(d)]\big) \tag{3}$$

The skip connection is critical: direct replacement $f_{\mathrm{adapt}} = \mathrm{MLP}(\cdot)$ would risk corrupting the semantic manifold if the MLP learns poor depth associations early in training. The residual formulation preserves the frozen semantic features as a strong prior while allowing the MLP to learn geometric adjustments. The MLP has one hidden layer with GELU activation—chosen over ReLU because ReLU's zero-gradient region stalls learning when depth features are sparse in planar regions like roads—and two-hundred-fifty-six hidden units, the smallest width that achieves saturation in validation mIoU on COCO.

DCFA is trained via single-epoch COCO self-distillation. COCO is chosen over ImageNet because COCO contains the instance-level diversity and spatial relationships (occlusion, adjacency, varying scales) that panoptic segmentation requires; ImageNet's single-object-centric images lack the compositional structure that depth-conditioned features must learn to disambiguate. One epoch is sufficient because DCFA is learning to correlate depth patterns with feature adjustments rather than learning a complex mapping from scratch; convergence monitoring shows no gain after the first epoch. Crucially, the DINOv3 backbone remains frozen throughout, preserving the self-supervised feature manifold.

SIMCF-ABC: resolving cross-modal inconsistency. The label-level error calls for a filter that resolves the coupled failures between semantic regions and depth instances. These failures are coupled: fixing fragmentation without fixing over-spill is futile, because merged fragments would still carry wrong labels. This coupling motivates a three-step filter where each step prepares the input for the next.

Step A enforces semantic majority voting within each instance mask: the majority semantic label is assigned to all pixels, preventing catastrophic over-spill into geometrically distinct objects. More sophisticated weighted voting—by pixel confidence or depth centrality—provides no measurable gain, so the simple majority scheme is retained.

Step B merges over-fragmented instances via DINOv3 cosine similarity computed on shared boundary pixels. The boundary-only computation is essential: computing similarity on full mask interiors would be dominated by background pixels (sky, road) that most instances share, making all objects appear similar. Boundaries capture the distinctive visual properties that separate adjacent objects. For masks M_i and M_j with shared boundary $\partial M_i \cap \partial M_j$, the similarity is:

$$s(i,j) = \cos\left(\frac{1}{|\partial M_i \cap \partial M_j|} \sum_{p \,\in\, \partial M_i \cap \partial M_j} f_{\mathrm{dino}}(p)\right) \tag{4}$$

Masks with $s(i,j) > \tau_{\mathrm{merge}}$ and identical semantic labels are merged. This step contributes $+1.48$ $\mathrm{PQ}_{\mathrm{th}}$—the single largest component gain in the pipeline—because the baseline produces approximately forty-four instances per image while the true instance count is approximately twelve.

Step C masks depth outliers: pixels whose depth deviates by more than three median absolute deviations from the instance median are masked as uncertain. The three-MAD threshold removes distant background pixels visible through object openings (e.g., sky through car windows) while preserving legitimate intra-object depth variation (e.g., a car hood is closer than its roof). Alternative thresholds (two MAD, four MAD) show no improvement; three MAD is robust to the long-tailed depth distributions typical of monocular depth estimators.

Stage 2: supervised training on pseudo-labels. With refined pseudo-labels from DCFA and SIMCF-ABC, we train a Cascade Mask R-CNN on one-hundred-percent Stage 1 pseudo-labels. The cascade architecture is chosen over standard Mask R-CNN because its multiple detection stages progressively refine bounding boxes, a property that is helpful when pseudo-labels are noisy at the box level—as they inevitably are from depth-guided splitting. The DINOv3 backbone remains frozen, integrated via a detectron2 adapter that projects patch features into the R-CNN pipeline without replacing the ResNet-50 FPN. DINOv3 features, while semantically rich, lack the explicit multi-scale hierarchy that FPN requires for object detection at varying scales; the ResNet-50 FPN provides this hierarchy from ImageNet pretraining, creating a hybrid design that gives the best of both domains—semantic understanding from DINOv3, multi-scale structure from ResNet-50.

Following the CUPS training protocol ensures fair comparison: eight-thousand steps, batch size four, DropLoss (p=0.5), CopyPaste augmentation. With baseline pseudo-labels, this achieves 31.62% PQ; with DCFA+SIMCF-ABC pseudo-labels, it reaches 33.29% PQ—a +1.67 PQ gain that is already significant, but not yet the full story.

Stage 3: EMA self-training and non-linear amplification. The final stage applies exponential moving average self-training with teacher momentum α=0.999:

$$\theta^{(t)}_{\text{teacher}} = \alpha\, \theta^{(t-1)}_{\text{teacher}} + (1-\alpha)\, \theta^{(t)}_{\text{student}} \tag{5}$$

for eight-thousand steps with an eighty-twenty mix of student-generated and teacher pseudo-labels.

The EMA formulation is chosen over hard teacher weight copying because it produces smoother pseudo-labels: rapid teacher updates introduce noise into the pseudo-label distribution, while EMA averaging stabilizes predictions. The high momentum means the teacher evolves slowly, preventing it from chasing the student's transient errors. The eighty-twenty ratio balances two failure modes: one-hundred-percent student labels cause drift (the student reinforces its own errors), while one-hundred-percent teacher labels prevent exploration (the student never sees hard examples the teacher mislabels).

The central observation of Stage 3 is that pseudo-label gains amplify non-linearly. A +1.31 PQ improvement in Stage 1 pseudo-labels becomes +4.21 PQ in the Stage 3 model—an amplification factor of 3.2×. This amplification reflects a structural property of EMA self-training: high-quality pseudo-labels provide consistent supervision across training steps, allowing the student to learn reliable decision boundaries, while low-quality pseudo-labels provide inconsistent supervision, causing the student to oscillate and converge to a poorer local minimum.

The amplification factor itself scales with teacher backbone quality: ResNet-50 (weak teacher, inconsistent pseudo-labels) amplifies by only 1.05× (+1.25 PQ), DINOv2 by 1.14× (+3.73 PQ), and DINOv3 by 1.29× (+7.96 PQ). This scaling behavior suggests that self-training is not merely a refinement stage but a phase transition: below a critical teacher quality threshold, self-training provides marginal gains; above the threshold, it exploits the structure of high-quality pseudo-labels to reach substantially better solutions.

With DCFA+SIMCF-ABC pseudo-labels and DINOv3 teacher, the final result reaches **35.83% PQ—+8.0 PQ above CUPS**—with thing-class quality improving from 27.37% to **36.26% $\mathbf{PQ_{th}}$**.

Suggested ablations. To strengthen the empirical foundation, we recommend the following ablations: k-sweep ($k \in \{20, 40, 50, 60, 80, 100, 120, 150\}$) to characterize the overclustering recovery curve; multiple random seeds (42, 123, 2024) to establish variance bounds; backbone sweep (ResNet-50, DINOv2 ViT-B/14, DINOv3 ViT-B/16) to isolate backbone contribution; depth estimator sweep (SPIdepth, DA2-Large, DA3 with fixed splitting) to quantify the depth quality binding constraint; splitting algorithm sweep (Sobel+CC, Canny, watershed, multiscale Sobel on fixed DA3 depth) to confirm diminishing returns from post-processing; oracle analysis (replace predicted person masks with ground truth) to quantify the person-class bottleneck; self-training scaling (vary teacher quality with fixed pseudo-labels) to establish the amplification law; and cross-dataset transfer (Mapillary Vistas v2 and KITTI without adaptation) to test domain invariance.